

Lab 07-Set Operations

Objective

Solving exercises from the textbook in chapter 2.1 & 2.2

Current Lab Learning Outcomes (LLO)

By completion of the lab, the students should be able to:

1. will understand set theory.
2. will be able to solve shorter/easier or longer / harder problems given in the textbook

Lab Requirements

Students allowed using their lecture notes in the lab in order to solve the exercises.

Lab Assessment

- 1- Divide students to groups and let them to solve the given example.
- 2- Discuss the answers with the groups and write on board the optimal solution.

Lab Description

- 1) List the members of these sets.
 - a) $\{x \mid x \text{ is a real number such that } x^2 = 1\}$
 - b) $\{x \mid x \text{ is a positive integer less than } 12\}$
 - c) $\{x \mid x \text{ is the square of an integer and } x < 100\}$
 - d) $\{x \mid x \text{ is an integer such that } x^2 = 2\}$
- 2) For each of these pairs of sets, determine whether the first is a subset of the second, the second is a subset of the first, or neither is a subset of the other.
 - a) the set of airline flights from New York to New Delhi, the set of nonstop airline flights from New York to New Delhi
 - b) the set of people who speak English, the set of people who speak Chinese
 - c) the set of flying squirrels, the set of living creatures that can fly
- 3) For each of the following sets, determine whether 2 is an element of that set.
 - a) $\{x \in \mathbf{R} \mid x \text{ is an integer greater than } 1\}$
 - b) $\{x \in \mathbf{R} \mid x \text{ is the square of an integer}\}$
 - c) $\{2, \{2\}\}$ d) $\{\{2\}, \{\{2\}\}\}$ e) $\{\{2\}, \{2, \{2\}\}\}$ f) $\{\{\{2\}\}\}$
- 4) Find the truth set of each of these predicates where the domain is the set of integers.
 - a) $P(x): x^2 < 3$
 - b) $Q(x): x^2 > x$
 - c) $R(x): 2x + 1 = 0$
- 5) Let A be the set of students who live within one mile of school and let B be the set of students who walk to classes. Describe the students in each of these sets.
 - a) $A \cap B$
 - b) $A \cup B$
 - c) $A - B$
 - d) $B - A$

- 6) Determine whether each of these statements is true or false.
a) $0 \in \emptyset$ **b)** $\emptyset \in \{0\}$ **c)** $\{0\} \subset \emptyset$ **d)** $\emptyset \subset \{0\}$
e) $\{0\} \in \{0\}$ **f)** $\{0\} \subset \{0\}$ **g)** $\{\emptyset\} \subseteq \{\emptyset\}$
- 7) What is the cardinality of each of these sets?
a) $\{a\}$ **b)** $\{\{a\}\}$ **c)** $\{a, \{a\}\}$ **d)** $\{a, \{a\}, \{a, \{a\}\}\}$
- 8) Find the power set of the sets where a and b are distinct elements
a) $\{a, b\}$ **b)** $\{\emptyset, \{\emptyset\}\}$
- 9) How many elements does each of these sets have where a and b are distinct elements?
a) $P(\{a, b, \{a, b\}\})$ **b)** $P(\{\emptyset, a, \{a\}, \{\{a\}\}\})$ **c)** $P(P(\emptyset))$
- 10) Let $A = \{a, b, c, d\}$ and $B = \{y, z\}$. Find
a) $A \times B$. **b)** $B \times A$.
- 11) What is the Cartesian product $A \times B \times C$, where A is the set of all airlines and B and C are both the set of all cities in the United States? Give an example of how this Cartesian product can be used.
- 12) Let $A = \{a, b, c, d, e\}$ and $B = \{a, b, c, d, e, f, g, h\}$. Find
a) $A \cup B$. **b)** $A \cap B$. **c)** $A - B$. **d)** $B - A$.
- 13) Let $A = \{0, 2, 4, 6, 8, 10\}$, $B = \{0, 1, 2, 3, 4, 5, 6\}$, Find $|A \cup B|$. find the minimum and maximum number of elements in $A \cup B$
- 14) Show that if A, B , and C are sets, then $\overline{A \cap B \cap C} = \overline{A} \cup \overline{B} \cup \overline{C}$
a) by showing each side is a subset of the other side.
b) using a membership table.
- 15) Let A, B , and C be sets. Show that
a) $(A \cup B) \subseteq (A \cup B \cup C)$. **b)** $(B - A) \cup (C - A) = (B \cup C) - A$.
- 16) Draw the Venn diagrams for each of these combinations of the sets A, B , and C .
a) $A \cap (B - C)$ **b)** $(A \cap B) \cup (A \cap C)$
c) $(A \cap \overline{B}) \cup (A \cap \overline{C})$
- 17) Show that if A is a subset of a universal set U , then
a) $A \oplus A = \emptyset$. **b)** $A \oplus \emptyset = A$. **c)** $A \oplus U = \overline{A}$. **d)** $A \oplus \overline{A} = U$.
- 18) Draw a Venn diagram that shows the following sample space and events:



S : all the integers from 1 to 30 P : prime numbers

M : multiples of 3 F : factors of 30